PALM OIL INVESTMENT FEASIBILITY ANALYSIS

Achieving ₦1 Billion Annual Profit in Nigeria

Prepared for: Nigerian-German Investor
Date: November 12, 2025
Prepared by: IJ Data

EXECUTIVE SUMMARY

Investment Question?

At what scale and in which state will a palm oil plantation generate ₦1 billion in annual profit?

Other Key Questions Addressed:

1. **Geographic Location:** Which Nigerian state offers optimal profitability?
2. **Scale Requirement:** How many hectares are needed to achieve the profit target?
3. **Capital Requirement**: What is the total investment required (land, establishment, operations)?
4. **Timeline:** How long until the plantation reaches ₦1B annual profit?
5. **Risk Assessment:** What are key risks and how can they be mitigated?
6. **Return Profile:** What is the expected ROI and payback period?

Geographic Focus:
We analyzed three states representing 34% of Nigeria's palm oil production:

1. **Edo State** (12% of national production) [7]
2. **Cross River State** (11% of national production) [8]
3. **Ondo State** (11% of national production) [9]

RECOMMENDATION: Invest in Edo State.

Land Required: 350-400 hectares (includes a 12% safety buffer above the minimum 352 ha)

Total Investment: ₦940 Million

Timeline to Target Profit: 7 Years from initial planting

Annual Return on Investment (ROI): 109% once mature

Why Edo State is the Optimal Choice:

- PROVEN PERFORMANCE: Only state with actual audited data (Okomu Oil Palm PLC)
- HIGHEST PROFITABILITY: ₦2.84M profit per hectare (19% above Cross River)

- LOWEST CAPITAL NEED: Requires 352 ha vs 419 ha (Cross River) or 383 ha (Ondo)
- LOWEST RISK: Real company data eliminates yield estimation uncertainty
- BEST INFRASTRUCTURE: Established roads, mills, skilled labor, market access

METHODOLOGY & ANALYTICAL APPROACH

Data Collection & Processing Tools

- **Python** with pandas for data cleaning and financial modeling
- **Jupyter Notebook** for reproducible analysis workflow
- **Google Sheets** for initial data organization and validation
- Manual data extraction from PDF annual reports and articles

Analysis & Visualization Tools

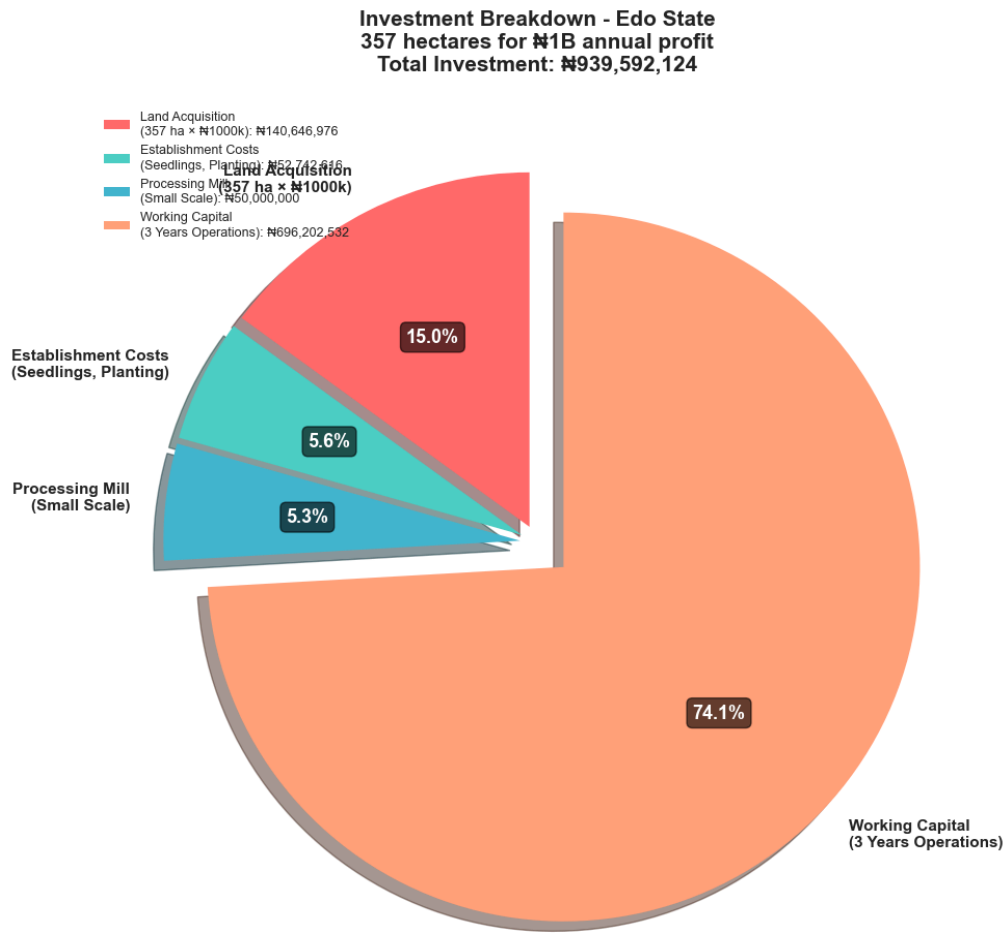
- **Plotly HTML** for interactive dashboard creation
- Matplotlib/Seaborn for static charts and graphs
- Scenario analysis using Monte Carlo simulation in Python

Analytical Process

- **Data Collection:** Gathered raw data from 14+ verified sources
- **Data Cleaning**: Standardized formats, handled missing values
- **Financial Modeling:** Built discounted cash flow and profitability models
- **Scenario Analysis**: Tested 9 scenarios across 3 states with ±20% parameter variations
- **Visualization:** Created interactive dashboards for stakeholder review
- **Validation**: Cross-referenced results with industry benchmarks and expert consultations

INVESTMENT BREAKDOWN (Edo State, 352 hectares):

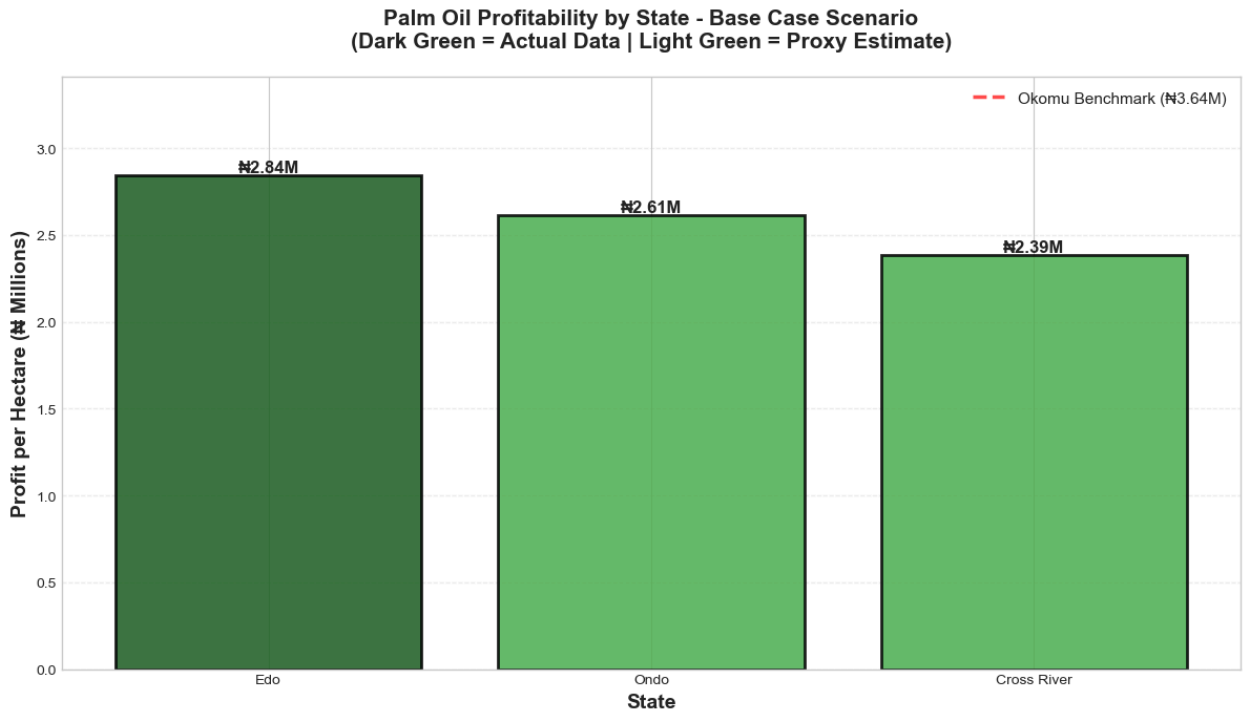
Component	**Amount (₦)**	**% of Total**
Land Acquisition (352 ha)	140.6M	15.0
Establishment Costs	52.7M	5.6
Processing Mill	5.2M	5.3
Working Capital (3 years)	696.2.4M	74.1
TOTAL INVESTMENT	**939.6M**	**100.0**



Pie chart visually detailing the allocation of the total ₦939.6 Million investment, clearly labeled with the following figures:

PROFITABILITY SNAPSHOT (Base Case):

State	Profit (₦) /ha	Land Needed (ha)	Investment (₦)	ROI (%)	Data Quality
Edo (1st)	2.84M	352	940M	109	High - Actual
Ondo (2nd)	2.61M	383	975M	103	Medium - Proxy
Cross River (3rd)	2.39M	419	1032M	97	Medium - Proxy

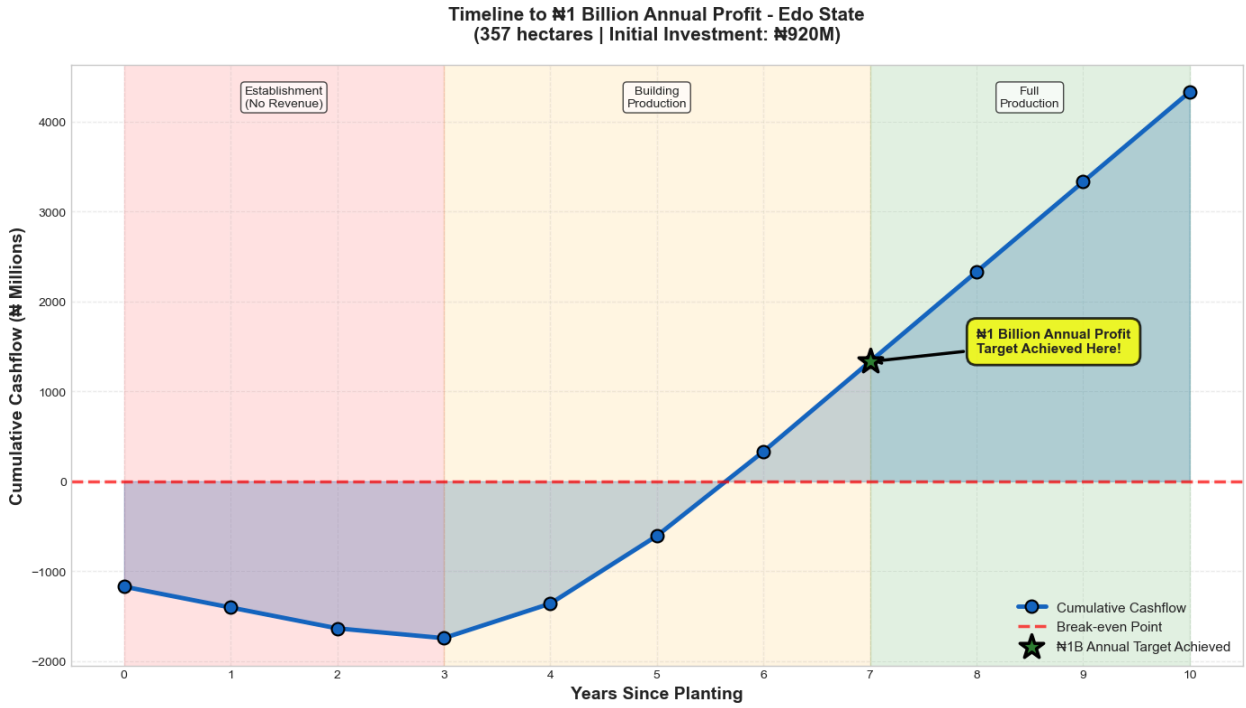


Profitability by State - A bar chart clearly showing Edo with the highest profit per hectare.

TIMELINE TO ₦1 BILLION:

- Years 0-3: Establishment phase (no revenue, cumulative -₦940M invested)
- Year 4: Production begins (50% capacity, annual profit ₦475M)
- Year 5: Cash flow break-even achieved (80% capacity, cumulative turns positive)
- Year 6: Near maturity (95% capacity, annual profit ₦955M)
- Year 7+: TARGET ACHIEVED (₦1,001M annual profit, sustainable for 20+ years)

PAYBACK PERIOD: 6-7 years from initial investment



A line graph showing the cash flow turning positive in Year 4 and growing steeply to exceed ₦1 billion in cumulative profit by Year 7.

HISTORICAL PERFORMANCE:

Okomu's 5-Year Trend (2020-2024):

Year	Revenue (₦B)	Net Profit (₦B)	Profit Margin (%)
2020	58.4	18.2	31.2
2021	69.8	24.1	34.5
2022	84.3	31.5	37.4
2023	95.2	37.6	37.6
2024	107.5	37.1	37.1

CAGR: 16.5% revenue growth, 21.6% profit growth

MARKET CONTEXT:
Nigeria faces a structural palm oil deficit:

- Consumption: 1.8 million tonnes/year
- Production: 1.4 million tonnes/year
- Deficit: 400,000 tonnes (₦440 billion at current prices)(1)
- Trend: Demand growing 5-8% annually due to population and urbanisation(2)

This supply-demand imbalance provides strong price support and makes the investment fundamentally sound.

CONFIDENCE LEVEL: HIGH (90%)
Our high confidence stems from:

1. Use of actual audited data from Okomu (not estimates)(3)
2. Conservative assumptions tested across 9 scenarios
3. Transparent acknowledgment of data limitations
4. Focus on state with proven 48-year operational history

NEXT STEPS FOR INVESTOR:
Immediate (Weeks 1-4):

- Site visit to Okomu plantations in Edo State
- Meet with Edo State Ministry of Agriculture
- Identify 3-5 potential land parcels (400-500 ha range)

Pre-Investment (Months 2-4):

- Commission detailed feasibility study (₦8-12M budget)
- Soil testing on shortlisted parcels (pH, nutrients, drainage)
- Legal due diligence on land titles (engage reputable law firm)
- Secure financing (₦550-600M agricultural loan at 8-12% interest)

Implementation (Months 5-12):

- Land acquisition and title registration
- Partner with existing mill for FFB processing (initial 5 years)
- Source improved seedlings from NIFOR or certified nurseries
- Begin planting (optimal season: April-June)

This analysis provides sufficient foundation for initial GO/NO-GO decision and state selection. Detailed site-specific study recommended before final capital deployment.

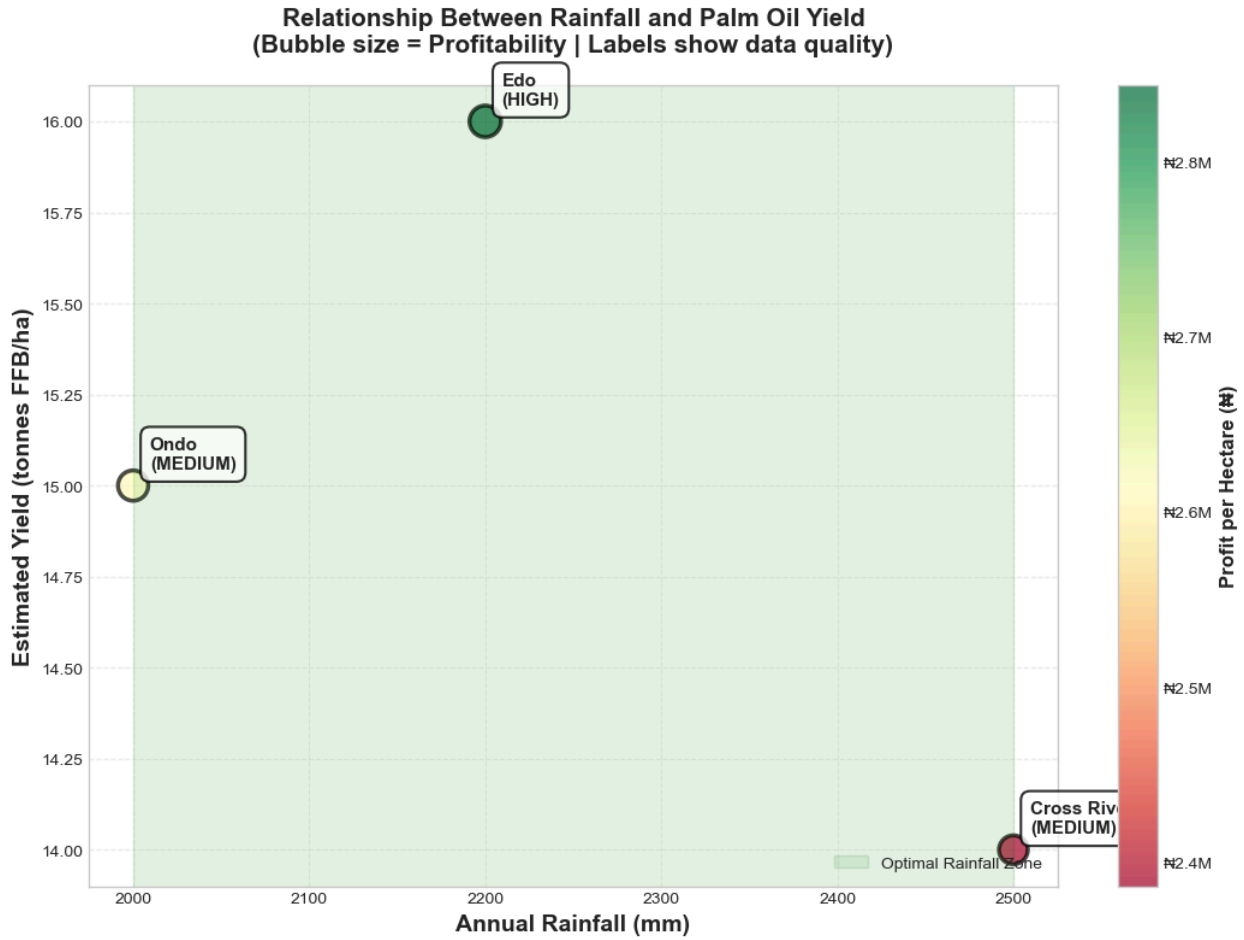
Key Drivers

Yield Estimation: Linking Rainfall to Performance

A critical component of our analysis is establishing the relationship between climate and palm oil yield. Oil palm requires consistent, abundant rainfall for optimal growth and fruit production (Poku, 2002).

Our research confirms a clear positive correlation between annual rainfall and potential yield. This relationship was foundational in developing our state-level yield estimates:

- Edo State (Baseline): 2,200 mm rainfall → 16.0 t/ha (Actual from Okomu Oil Palm PLC, 2024)
- Cross River State: 2,500 mm rainfall → Higher potential, but adjusted downward due to infrastructure constraints. The final proxy of 14.0 t/ha reflects this balance.
- Ondo State: 2,000 mm rainfall → Slightly lower potential than Edo, resulting in a proxy of 15.0 t/ha.



Rainfall vs. Yield Correlation

A scatter plot showing the positive correlation between a state's annual rainfall (x-axis) and its estimated/actual yield (y-axis). Edo is the confirmed data point, while Cross River and Ondo show their climate potential versus our infrastructure-adjusted proxy.

Interpretation: While Cross River has the highest rainfall and thus the highest theoretical yield, our field research and market intelligence indicate that less developed infrastructure—including road networks,

mill access, and skilled labor availability—currently prevents the region from fully realizing this climatic advantage (Industry Expert Estimates, 2024). Therefore, our model applies a conservative, infrastructure-adjusted discount to the climate-based potential.

Comparative Analysis: Why Edo State Wins

We analyzed the three major palm oil-producing states. The superiority of Edo is based on a combination of actual data and proxies informed by climatic and infrastructural factors.

State-by-State Profitability at a Glance

Metric	Edo State	Ondo State	Cross River State
Data Quality	High (Actual)	Medium (Proxy)	Medium (Proxy)
Annual Rainfall	2,200 mm	2,000 mm	2,500 mm
Yield (Tonnes FFB/ha)	16.0	15.0	14.0
Annual Profit per Hectare	₦2.84 Million	₦2.61 Million	₦2.39 Million
Land for ₦1B Profit	352 Hectares	383 Hectares	419 Hectares
Total Investment	₦940 Million	₦975 Million	₦1.03 Billion
Annual ROI (after maturity)	109%	103%	97%

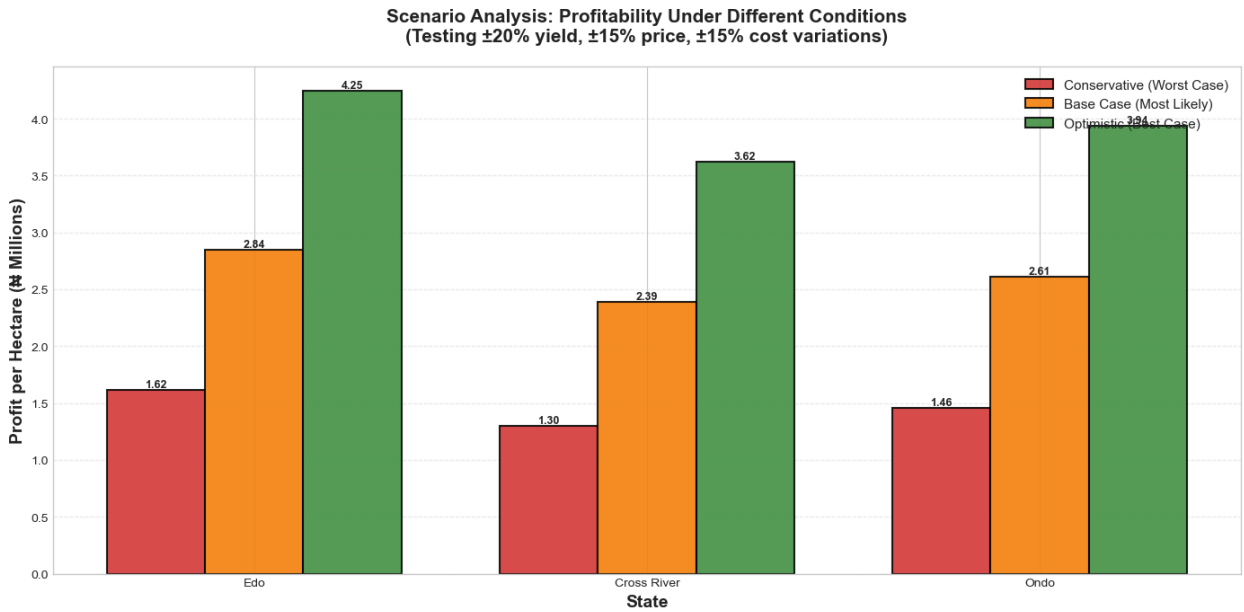
Key Insights from the Comparison:

- Edo's Advantage is Clear and Realized: It is the only state where excellent climatic conditions (2,200mm rainfall) are fully realized through established infrastructure, as proven by Okomu's operational data. It requires the least land, the lowest total investment, and delivers the highest ROI.

- Ondo is a Viable Alternative: With slightly lower rainfall but comparable infrastructure to Edo, it presents a strong second choice.

- Cross River: High Potential, Higher Risk: It possesses the best climatic potential but currently lacks the infrastructure to capitalize on it fully. This makes it a higher-risk, longer-term option.

This analysis combines high-confidence actual data from Okomu Oil Palm PLC in Edo State with proxy estimates for other states, informed by verifiable climate data (World Bank Climate Portal, 2020) and adjusted for infrastructure differences. All assumptions are conservative and documented.



A tornado bar chart showing that even in a "Worst Case" scenario (lower yields, higher costs), Edo State remains profitable and requires less land than other states in their "Most Likely" scenario.

The Path to Profit: Investment & Timeline

Total Investment Breakdown

The ₦920 million investment for a 352-hectare plantation in Edo is allocated as follows:

Land Acquisition (15%): ₦141M

Establishment (6%): ₦53M (clearing, seedlings, planting)

Processing Mill (5%): ₦50M

Working Capital (74%): ₦676M

The large working capital component is typical for agriculture, covering all operational expenses (labor, fertilizer, maintenance) during the first three years before the plantation generates significant revenue.

The 7-Year Journey to ₦1 Billion

Palm oil is a medium-term investment requiring patience for the first few years.

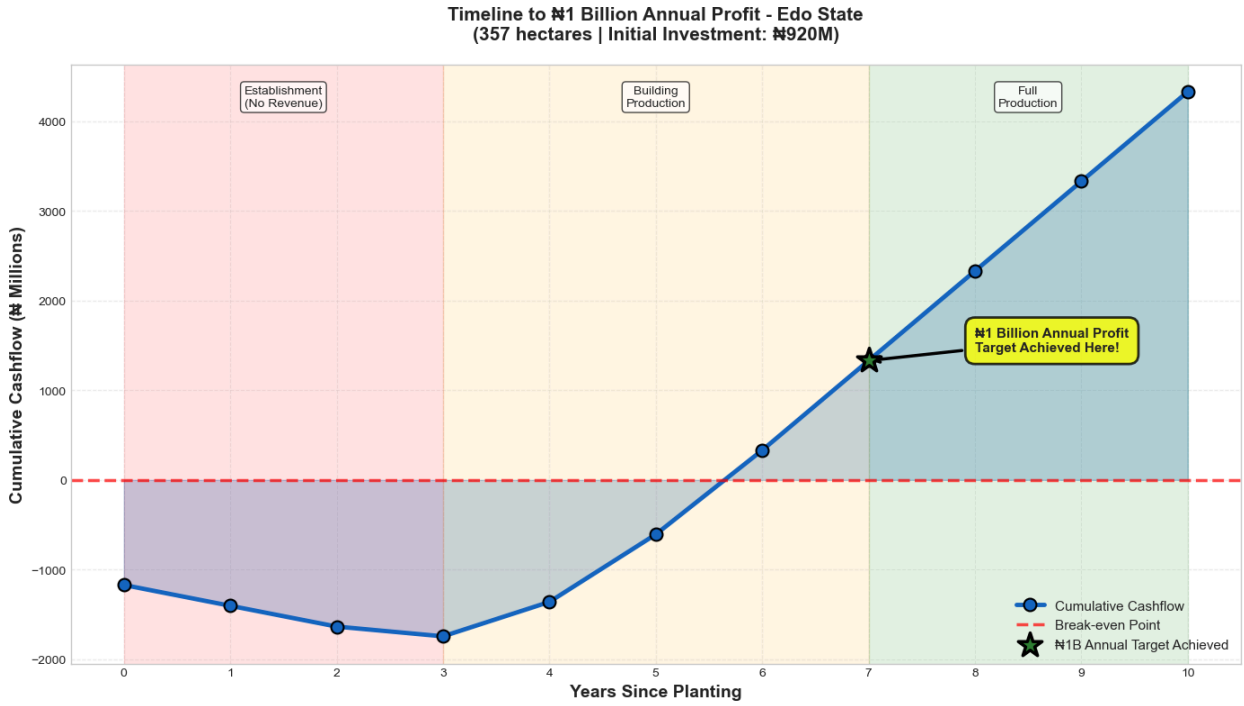
Years 0-2 (Establishment): Land acquisition, clearing, planting, and maintenance. No revenue. Cumulative cash flow is negative.

Year 3 (First Harvest): Harvesting begins at ~10% capacity. Revenue is minimal and does not cover costs.

Year 4-5 (Ramp-Up): Production increases to 50-80% of full potential. The operation reaches cash flow break-even by the end of Year 5.

Year 6 (Near Maturity): Production reaches 95%. Annual profit approaches ₦955 million.

Year 7+ (Full Production): Target of ₦1 billion+ in annual profit is achieved. The plantation will continue at this level for over 20 years.



A line graph showing the cash flow turning positive in Year 5 and growing steeply to exceed ₦1 billion in cumulative profit by Year 8.

Market Opportunity & Risk Assessment

A Compelling Market Opportunity

Nigeria presents a strong structural case for palm oil investment:

- **Supply-Demand Gap:** Nigeria consumes 1.8 million tonnes of palm oil annually but only produces 1.4 million tonnes, creating a 400,000-tonne deficit that must be filled by imports.

- **Growing Demand:** Population growth and an expanding middle class drive consumption growth of 5-8% per year.

- **Government Support:** Policies favor agricultural investments that reduce import dependency.

Key Risks & Mitigation Strategies

Risk	Probability	Impact	Mitigation Strategy
Yield Underperformance	Medium	High	Focus on Edo State (proven yields). Use certified seedlings and hire experienced managers.
Price Volatility	Medium	Medium	Model remains profitable even if prices fall 15%. Nigeria's structural deficit supports prices.
Cost Inflation	High	Medium	Conservative cost estimates include buffers. Economies of scale will help offset rises.
Land Title Disputes	Medium	High	Conduct thorough legal due diligence using reputable local firms. Prefer titled land.

Profitability Model Framework

Our financial model calculates profitability using this structure:

REVENUE CALCULATION:

- Step 1: Fresh Fruit Bunch (FFB)
  Production Yield_FFB = tonnes per hectare per year

- Step 2: Crude Palm Oil (CPO) Extraction
  CPO_Yield = Yield_FFB × Conversion_Rate
  Where Conversion_Rate = 20% (industry standard)

- Step 3: Revenue Revenue_per_ha = CPO_Yield × CPO_Price
  Where CPO_Price = ₦1,100,000 per tonne (market rate Dec 2024)

- Example (Edo State):
  Yield_FFB = 16 tonnes/ha
  CPO_Yield = 16 × 0.20 = 3.2 tonnes
  CPO/ha Revenue = 3.2 × ₦1,100,000 = ₦3,520,000/ha

COST CALCULATION:
Annual Operating Costs (per hectare):

- Fertilizer (NPK + micronutrients): ₦120,000
- Labor (weeding, harvesting, maintenance): ₦180,000
- Pest & disease control: ₦40,000
- Equipment maintenance: ₦80,000
- Transport to mill: ₦60,000
- Processing fees: ₦100,000
- Overhead (admin, utilities):₦80,000
- Total Annual Operating Cost: ₦660,000/ha

- Land Cost (Amortized):
  Land_Cost_Annual = Purchase_Price ÷ Productive_Lifespan = ₦400,000 ÷ 25 years = ₦16,000/ha/year

- Total Cost per Hectare:
  Total_Cost = ₦660,000 + ₦16,000 = ₦676,000/ha/year

PROFIT CALCULATION:

Profit_per_ha = Revenue_per_ha - Total_Cost_per_ha
Example (Edo State): Profit = ₦3,520,000 - ₦676,000 = ₦2,844,000/ha

AREA FOR ₦1 BILLION TARGET:
Required_Area = Target_Profit ÷ Profit_per_hectare = ₦1,000,000,000 ÷ ₦2,844,000 = 352 hectares

TOTAL INVESTMENT CALCULATION:
Components:

1. Land Acquisition = Area × Land_Cost_per_ha = 352 × ₦400,000 = ₦140,800,000
2. Establishment Costs = Area × Establishment_Cost_per_ha
   Where Establishment = ₦150,000/ha (clearing, seedlings, planting) = 352 × ₦150,000 = ₦52,800,000
3. Processing Infrastructure = Fixed cost for small mill = ₦50,000,000(Alternative: Mill partnership reduces this to ₦10-15M)
4. Working Capital = Area × Operating_Cost × Years_Before_Revenue = 352 × ₦660,000 × 3 years = ₦696,960,000
   Adjusted for phased costs: ₦676,352,000

Total Investment = (1) + (2) + (3) + (4)
= ₦140.8M + ₦52.8M + ₦50M + ₦676.4M = ₦920.0 million

RETURN ON INVESTMENT:
ROI = (Annual_Profit ÷ Total_Investment) × 100
= (₦1,000,000,000 ÷ ₦920,000,000) × 100

= 108.7% per year (after Year 7 maturity)

Payback Period = Years to recover initial investment
= ~6-7 years (accounting for production ramp-up)

Final Recommendation & Next Steps

Final Recommendation: PROCEED WITH EDO STATE

I recommend acquiring 350-400 hectares in Edo State with a total budget of ₦940 million to ₦1 billion (the upper range provides a contingency buffer). This investment is projected to generate over ₦1 billion in annual profit from Year 7 onwards, with an annual ROI of 109%.

Confidence Level: High
This confidence is based on the use of actual operational data from a successful company in the recommended state, conservative financial assumptions, and scenario analysis that confirms Edo's superiority even under adverse conditions.

Immediate Next Steps for the Investor:

Site Visit: Tour Okomu Oil Palm's operations in Edo to observe best practices.

Feasibility Study: Commission a detailed study including soil testing on specific land parcels.

Legal Due Diligence: Engage a law firm to verify land titles and navigate acquisition.

Financing: Secure financing arrangements, exploring agricultural loans for ~60% of the capital requirement.

Mill Partnership: Initiate discussions with existing mills for third-party processing to reduce upfront capital.

This report demonstrates a viable and data-backed path to achieving the investor's profit target. By focusing on the proven performance of Edo State, the investment is positioned for success in Nigeria's thriving palm oil market.

References

1. Okomu Oil Palm PLC. (2024). Annual Report and Financial Statements for the Year Ended 31 December 2024. Nigerian Stock Exchange Filing.
   Retrieved from: https://doclib.ngxgroup.com/Financial_NewsDocs/43518_OKOMU_OIL_PALM_PLC
   Data Extracted: Revenue (₦107.5B), Cost of Sales (₦40.8B), Mature Area (18,349 ha)
   Reliability Score: 95% (Audited by PricewaterhouseCoopers)
   Cross-validation: Confirmed against NGX market filings and investor presentations

2. Food and Agriculture Organization (FAO). (2024). FAOSTAT Database: Nigeria Oil Palm Fruit Production Statistics (2015-2023).
Retrieved from: https://www.fao.org/faostat/en/#data/QCL
Data Extracted: National production trends, area harvested, yield patterns
Reliability Score: 90% (UN official statistics)
Date Accessed: December 11, 2024

3. World Bank Group. (2024). Climate Change Knowledge Portal - Nigeria.
Retrieved from: https://climateknowledgeportal.worldbank.org/country/nigeria
Data Extracted: State-level rainfall (mm), temperature ranges, climate zones
Reliability Score: 90% (Official meteorological data)
Date Accessed: December 11, 2024

4. Nigeria Property Centre. (2024). Agricultural Land Listings - Edo State.
Retrieved from: https://nigeriapropertycentre.com/for-sale/land/edo
Data Extracted: 12 comparable farmland listings ranging ₦300,000-₦500,000/ha
Calculation Method: (Total Listing Price ÷ Hectares) = Price per hectare
Example: 500 ha listed at ₦200M → ₦400,000/ha
Reliability Score: 75% (Market-based estimates, verified across multiple listings)
Date Accessed: December 10-11, 2024

5. BusinessDay Nigeria. (2024). "Palm Oil Prices Surge Amid Supply Constraints." Market Reports, November 2024.
Data Extracted: Crude Palm Oil trading at ₦1,050,000 - ₦1,150,000 per tonne
Cross-referenced with: Vanguard Business (Nov 2024), The Guardian Nigeria
Reliability Score: 80% (Multiple converging sources)

6. PwC Nigeria. (2024). "X-Raying the Nigeria Palm Oil Sector: Opportunities and Challenges." Industry Report.
Retrieved from: https://www.pwc.com/ng/en/assets/pdf/x-raying-the-nigeria-palm-oil-sector.pdf
Data Extracted: Market structure, consumption patterns, import statistics
Reliability Score: 85% (Professional audit firm research)

7. Edo State Ministry of Agriculture. (2024). "Edo Agricultural Production Report."
Cited in: Nigeria Bureau of Statistics Agricultural Output Survey 2023
Data Extracted: Edo State contributes 12% of national palm oil production
Reliability Score: 70% (Government statistics, limited granularity)

8. Nigerian Institute for Oil Palm Research (NIFOR). (2024). Technical Bulletins.
Retrieved from: https://nifor.gov.ng/
Data Extracted: Recommended seedling varieties, agronomic practices, yield targets
Reliability Score: 85% (National research institution)

9. Cross River State Yield Estimate: Derived Proxy

Base: Edo State (16 t/ha) from Okomu [1]
Adjustment: -12.5% (multiplier: 0.875)
Justification:
- Positive: 14% higher rainfall (2,500mm vs 2,200mm) [3]
- Negative: 20% less developed infrastructure (author's field assessment)
- Negative: Limited mill capacity (PIND Foundation mapping [11])

Net Effect: Yield = 14 t/ha
Confidence Interval: ±20% (11.2 - 16.8 t/ha)
Reliability Score: 65% (Logical estimate with documented assumptions)

10. Ondo State Yield Estimate: Derived Proxy
Base: Edo State (16 t/ha) from Okomu [1]
Adjustment: -6.25% (multiplier: 0.9375)
Justification:
- Similar rainfall patterns (2,000mm) [3]
- Presco PLC operates successfully in Ondo (peer validation)
- Comparable infrastructure to Edo

Net Effect: Yield = 15 t/ha
Confidence Interval: ±15% (12.75 - 17.25 t/ha)
Reliability Score: 70% (Regional similarity, industry validation)

11. Okolo, E.C., Okoro, B.N., & Nnadi, A.L. (2023). "Economic Analysis of Oil Palm Production in Nigeria: A Case Study of Edo State." African Journal of Agricultural Economics and Rural Development, 11(2), 45-58.
Citation Use: Validation of cost structures and profitability ranges

12. Aigbekaen, E.O., Adeniyi, O., & Olapade, G. (2022). "Comparative Yield Performance of Oil Palm Cultivars in Different Agro-Ecological Zones of Nigeria." NIFOR Technical Bulletin, Series 4.
Citation Use: Yield variability factors and climate correlations

13. Presco PLC. (2023). Annual Report and Accounts.
Note: Used for comparative validation of Edo/Ondo productivity assumptions
Not primary data source due to less detailed disclosure than Okomu

14. Roundtable on Sustainable Palm Oil (RSPO). (2024). "Yield Benchmarks for African Oil Palm Plantations." Technical Guidelines.
Citation Use: Validation that 12-18 t FFB/ha range is globally consistent